PES2UG24CS118 Bhuvi Dange Section B
Lab 3: Component Modelling & Architectural Pattern Selection
Written Justification - Resident Equipment Borrowing System

Architecture Selection
We chose Layered Architecture for the Resident Equipment Borrowing System.

Reason 1
The system naturally splits into three concerns that map directly onto layers: a Presentation layer for resident and administrator interaction (registration, browsing equipment, submitting requests, approvals), a Business layer that owns the borrowing rules (eligibility checking, issuing, returns, repair fees, notifications), and a Data layer that persists equipment and resident records. Keeping the eligibility and approval logic centralized in one Business layer component makes it easy to reason about and test independently of the UI or the storage technology used underneath.

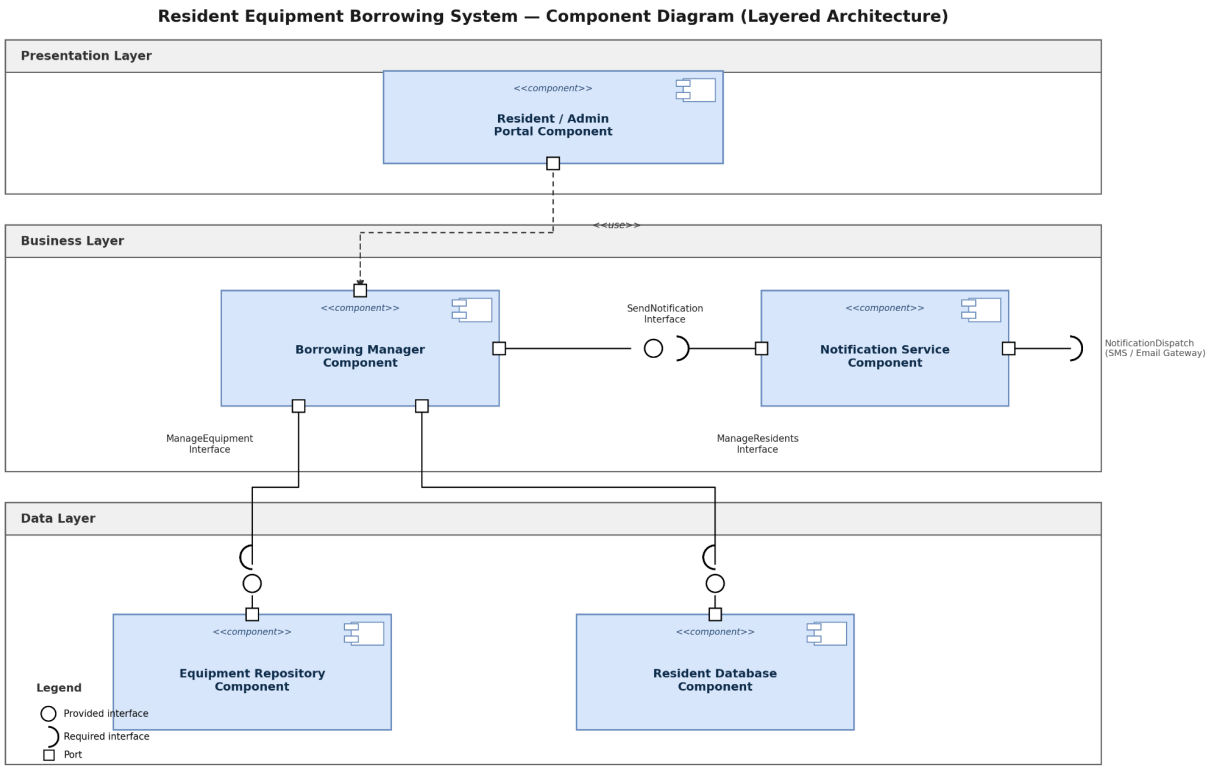
Reason 2
The functional scope is moderate -a single Borrowing Manager governs registration, requests, approvals, issuing and returns rather than dozens of independently-scaling capabilities. The operational overhead of a Microservices split (service discovery, distributed data consistency, network latency between services) is not justified here, while a Layered design still lets peripheral concerns, such as the Notification Service, be swapped or extended (e.g., moving from email to SMS reminders) without touching the core borrowing rules.

Security Advantage
NFR-02 requires that approving borrowing requests, recording issue/return, and charging repair fees be restricted to authenticated administrator accounts only. In a layered design this authorization check is enforced once, centrally, in the Business layer (inside the Borrowing Manager Component) before any request is allowed to reach the Data layer. This gives the system a single, well-defined choke point for access control, so a resident session can never bypass the UI and directly manipulate equipment availability or fee records in the Data layer.

Performance Benefit
FR-03 requires that a borrowing request be validated against two conditions - equipment availability and the resident's overdue status - and FR-01 requires registration to confirm within 2 seconds. Because both checks are performed inside the same Business layer component, the Borrowing Manager can query the Equipment Repository and Resident Database directly as local/in-process calls rather than making remote calls across independently deployed services. This keeps the request path short and predictable, helping the system meet its response-time targets without the added network latency that a Microservices architecture would introduce for such a tightly-coupled operation.

Components (Component Diagram)

1. Resident / Admin Portal Component (Presentation layer) - handles touch/web UI interactions for registration, browsing equipment, submitting/reserving requests, and admin approvals.
2. Borrowing Manager Component (Business layer) – core logic: registration processing, eligibility checking (availability + overdue), request approval, issue/return recording, and repair-fee charging.
3. Notification Service Component (Business layer) – sends return-due reminders and equipment-availability alerts to residents.
4. Equipment Repository Component (Data layer) – stores equipment records, category, condition, and availability status.
5. Resident Database Component (Data layer) – stores resident accounts, borrowing history, overdue status, and repair-fee charges.



Key Interfaces
- Portal Component <<use>> Borrowing Manager Component
- Borrowing Manager Component - Equipment Repository Component: ManageEquipment Interface (check/update availability)
- Borrowing Manager Component - Resident Database Component: ManageResidents Interface (check overdue status, update account/fee records)
- Borrowing Manager Component - Notification Service Component: SendNotification Interface (trigger reminders and availability alerts)
- Notification Service Component - NotificationDispatch (required interface to an external SMS/Email gateway)